

$$\begin{array}{l}
 X^{(1)T} \rightarrow \\
 X^{(2)T} \rightarrow \\
 \vdots \\
 X^{(m)T} \rightarrow
 \end{array}
 \begin{array}{c}
 \begin{matrix} a_1 & a_2 & & a_n \\ (x_1^{(1)} & x_2^{(1)} & \dots & x_n^{(1)} \\ (x_1^{(2)} & x_2^{(2)} & \dots & x_n^{(2)} \\ \vdots & \vdots & & \vdots \\ (x_1^{(m)} & x_2^{(m)} & \dots & x_n^{(m)} \end{matrix}
 \end{array}
 \begin{array}{c}
 \begin{matrix} y^{(1)} \\ y^{(2)} \\ \vdots \\ y^{(m)} \end{matrix}
 \end{array}$$

$X$   
 $(m, n)$

$y$   
 $(m, 1)$

$$X^{(i)} \in \mathbb{R}^n$$

$$y^{(i)} \in \mathbb{R}$$

$$D = \{(x^{(1)}, y^{(1)}), \dots, (x^{(m)}, y^{(m)})\}$$

Si un problema es en los reales es regresión  
si no clasificación

$$h_\theta(x) = w_1 x_1 + w_2 x_2 + \dots + w_n x_n + b = \sum_{j=1}^n w_j x_j + b = w^T x + b = x^T w + b$$

$b = \text{El sesgo}$

Si  $x = (x_1, \dots, x_n) \in \mathbb{R}^n$   
 $w = (w_1, \dots, w_n) \in \mathbb{R}^n$

↑ todos los parámetros

$$\theta = (w_1, \dots, w_n, b) \in \mathbb{R}^{n+1}$$

Aprendizaje

$$w^* = \text{mejor } w$$

$$\checkmark E_{in}(h^*) \approx E_{out}(h^*)$$

$$E_{in}(h^*) \approx 0$$

$$h^* = \arg \min_{h \in H} E_{in}(h) \leftrightarrow \theta^* = w^*, b^* = \arg \min_{\substack{w \in \mathbb{R}^n \\ b \in \mathbb{R}}} E_{in}(h_{w,b})$$

$$\frac{1}{m} = \text{promedio}$$

$$w^*, b^* = \arg \min_{\substack{w \in \mathbb{R}^n \\ b \in \mathbb{R}}} \frac{1}{m} \sum_{i=1}^m \frac{1}{2} (y^{(i)} - h_{w,b}(x^{(i)}))^2$$

MSE  
Mean, Square Error

$$w^*, b^* = \arg \min_{\substack{w \in \mathbb{R}^n \\ b \in \mathbb{R}}} \frac{1}{m} \sum_{i=1}^m \frac{1}{2} (y^{(i)} - x^{(i)T} w - b)^2$$

$$\arg \min_{\substack{w \in \mathbb{R}^n \\ b \in \mathbb{R}}} \frac{1}{2m} \left[ y^{(1)} - x^{(1)T} w - b \right] \cdots \left[ y^{(m)} - x^{(m)T} w - b \right]$$

$$\begin{bmatrix} y^{(1)} - x^{(1)T} w - b \\ y^{(2)} - x^{(2)T} w - b \\ \vdots \\ y^{(m)} - x^{(m)T} w - b \end{bmatrix}$$

$$\arg \min_{\theta \in \mathbb{R}^{n+1}} \frac{1}{2m} [Y - [X, \vec{1}] \theta]^T [Y - [X, \vec{1}] \theta]$$

$$\begin{bmatrix} y^{(1)} \\ y^{(2)} \\ \vdots \\ y^{(m)} \end{bmatrix} \begin{bmatrix} x^{(1)} \\ x^{(2)} \\ \vdots \\ x^{(m)} \end{bmatrix}^T \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} = b$$

$$F(x_1, \dots, x_n)$$

$$F(x) \quad x \in \mathbb{R}^n, F: \mathbb{R}^n \rightarrow \mathbb{R}$$

$$\nabla_x F(x) = \begin{bmatrix} \frac{\partial F(x)}{\partial x_1} \\ \frac{\partial F(x)}{\partial x_2} \\ \vdots \\ \frac{\partial F(x)}{\partial x_n} \end{bmatrix}$$

derivadas

$$\frac{\partial (V(x)^2)}{\partial x} = 2V(x) \frac{\partial V(x)}{\partial x}$$

$$\frac{1}{2m} [Y - X_e \theta]^T [Y - X_e \theta]$$

$$X_e^T \frac{1}{2m} [Y - X_e \theta]$$

$(n+1, m) \quad m, 1$

$$X_e^T y - X_e^T X_e \theta = 0 \Rightarrow$$

$$X_e^T X_e \theta = X_e^T y$$

$$\theta \equiv \underbrace{[X_e^T X_e]^{-1} X_e^T}_{\text{pinv}(X_e)} y$$

pinv( $X_e$ )

Moore-Penrose pseudoinverse

$X_e = X$  extended